

QIC 890: Quantum Complexity Theory

Guidelines for Problem Sets

Goals of the problem sets

Solving problems is a critically important part of the learning process, and this is the principal goal of the problem sets — to aid in learning by providing problems that are challenging but within reach.

Secondarily, the problem sets are an assessment tool. It is impossible for a problem set to assess your *ability* to solve problems, as AI is capable of finding solutions to any problem that could reasonably be asked in this course. Instead, the aim is to assess your ability to *explain solutions*, clearly and effectively. When preparing solutions you should therefore focus on clarity and effectiveness (as well as correctness, of course).

Target audience and level of detail

Write your solutions for a hypothetical classmate who has read and understood the course notes, but who has not attempted to solve the problems. The course notes are a reasonable target for the level of detail that is appropriate.

- Low-level details that are routine and add nothing to clarity are probably best skipped. (E.g., a high-level, plain-language description of a Turing machine computation is much more illuminating than a low-level, technical description of its transition function.)
- On the other hand, you should not dismiss important aspects of solutions just because they are easy. “Proof by intimidation” is not a valid proof technique; if something is truly easy, then it should be easy to explain.

Typical problem solutions are expected to require somewhere between half a page (for simpler problems) and two pages (for more complicated ones), though it is fine to deviate from this if a solution calls for it. There is generally no single ideal answer, so use your best judgment based on what you think a clear answer requires.

Sources, collaboration, and AI

- You are encouraged to try to solve the problems on your own. Struggling with challenging problems is great practice for any researcher, and also aids in understanding pitfalls and developing sensitivity to the reader.
- Discussing the problems with your classmates is allowed, provided you acknowledge those discussions. Likewise, use of AI tools is allowed, but must be honestly acknowledged. You will not lose points for acknowledging collaborations or use of AI, though losing points is possible if you fail to make these acknowledgments.
- In all cases, the solutions you submit must be written by you, in your own words. In particular, do not ask AI to write your solutions for you! (A typical problem solution written by AI is — at least for now — likely to be below the standard expected for this course.)
- You may use any source you like, but anything other than the course notes must be cited (including books, papers, lecture notes from other courses, and websites). Facts proved in the notes do not need to be proved again — but please refer to them explicitly (e.g., by theorem or equation number).

Logistics

- Email solutions as a single PDF attachment to watrous@uwaterloo.ca. Please use the subject “QIC 890 problem set N solutions” (with N replaced by 1, 2, 3, etc.).
- Solutions are due at 11:59 pm Eastern Time on the date given on each problem set.
- Late submissions received within 48 hours of the deadline will be penalized 25%, after which submissions will not be accepted.
- Solutions typeset in $\text{\LaTeX}$ are strongly encouraged and greatly appreciated.